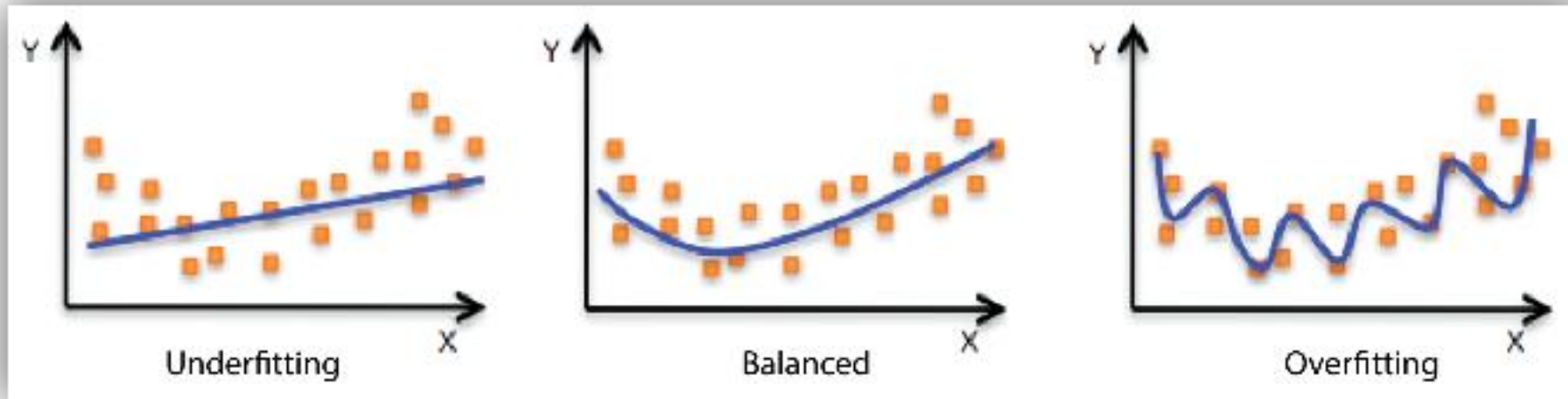


Model fitting challenges

Understanding model fit



Underfitting/Overfitting is one of the main causes for poor model accuracy

Underfitting

Not modelling the data well enough!

- Inadequate learning while training for the exam
- Consequently, bad performance in the exam

HOW I STUDY.



SLOTHILDA.COM

Overfitting

Fitting data at all costs!



Amitabh Bachchan ✓
@SrBachchan



T 3707 - What is the date today ..
31 . 10 . 20 .. add up ➡ .. 3 + 1 is 4 .. then 1 = 5 .. then 0 ..
then 2, so 4+1+2 = 7 .. then 0 ..
so .. 3+1+1+2 = 7 .. and 2 zeros before you get there ..
So .. 007 .. !!
Sean Connery passes away .. he gave life to 007 !!

10:45 PM · Oct 31, 2020



22K

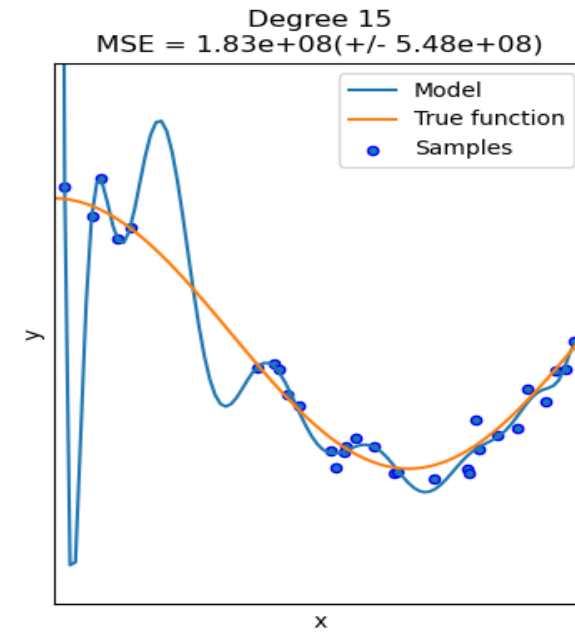
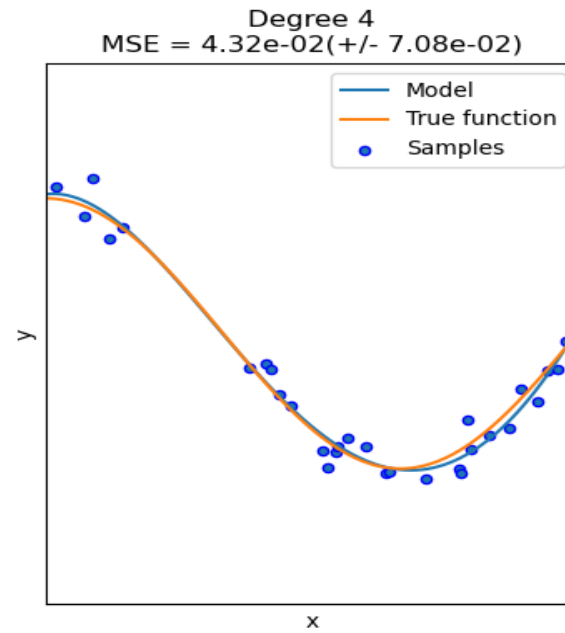
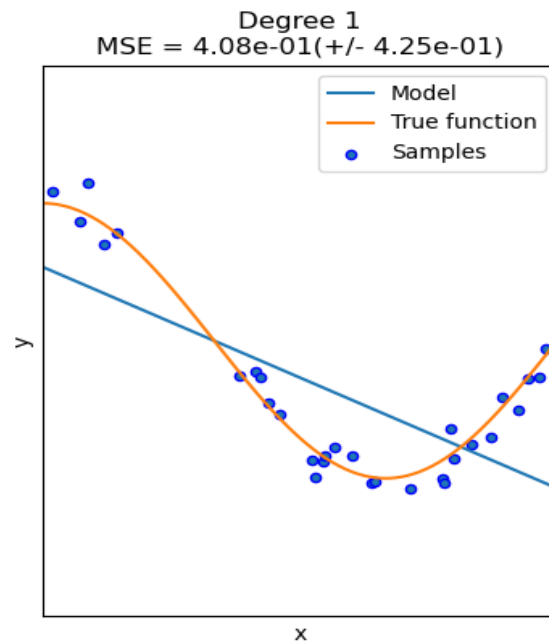


4.8K people are Tweeting about this

Overfitting

$$y = \beta_0 + \beta_1 x$$

$$y = \beta_0 + \beta_1 x + \beta_2 x^2 + \beta_3 x^3 + \dots + \beta_n x^n$$



Overfitting

linear regression (polynomial degree = 1)

linear regression with higher degree
polynomial features to approximate
nonlinear functions

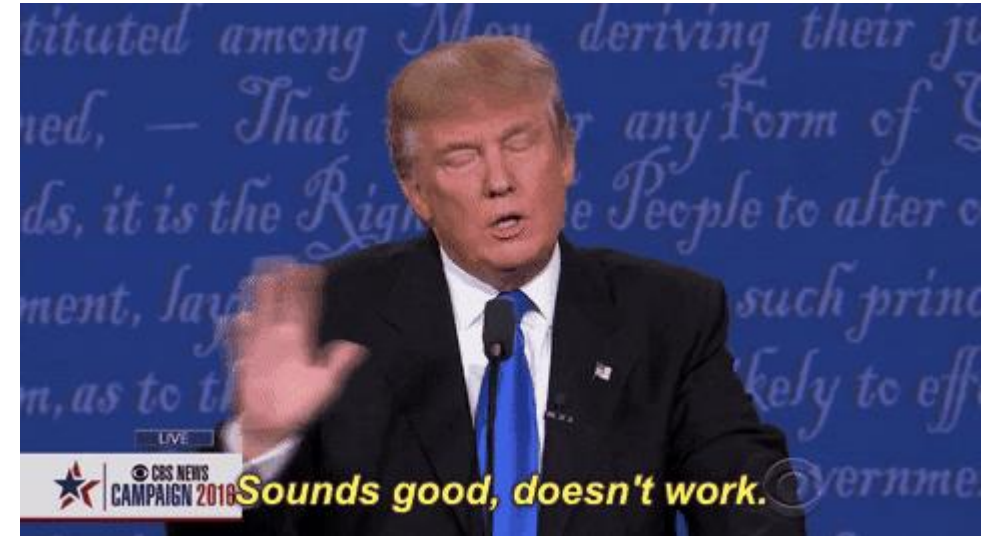
Detecting Overfitting

The key indicator

TRAINING ACCURACY

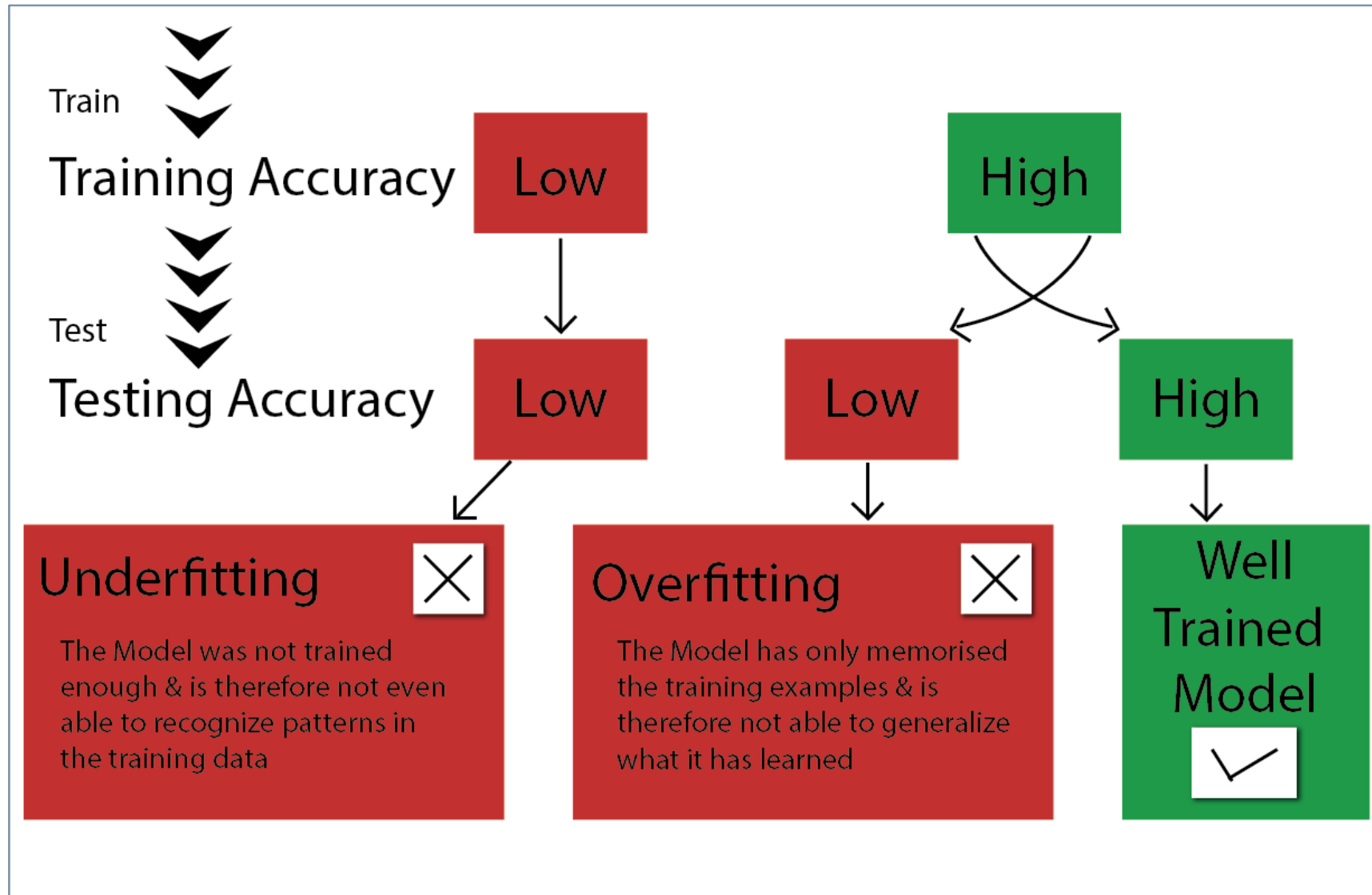


TESTING ACCURACY



Detecting Underfitting/Overfitting

Summary



Reasons for Underfitting

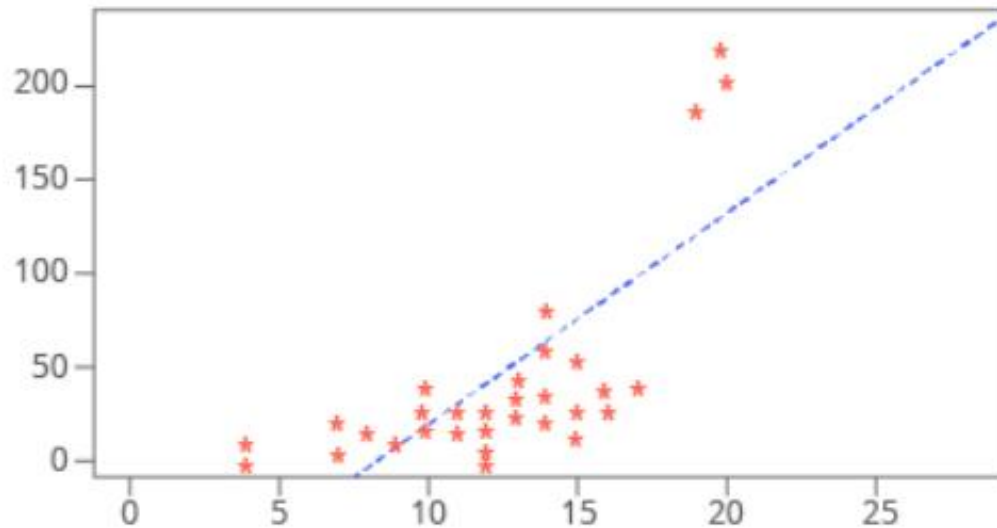
- Short training
- Simplistic model
- Lack of enough features

Reasons for Overfitting

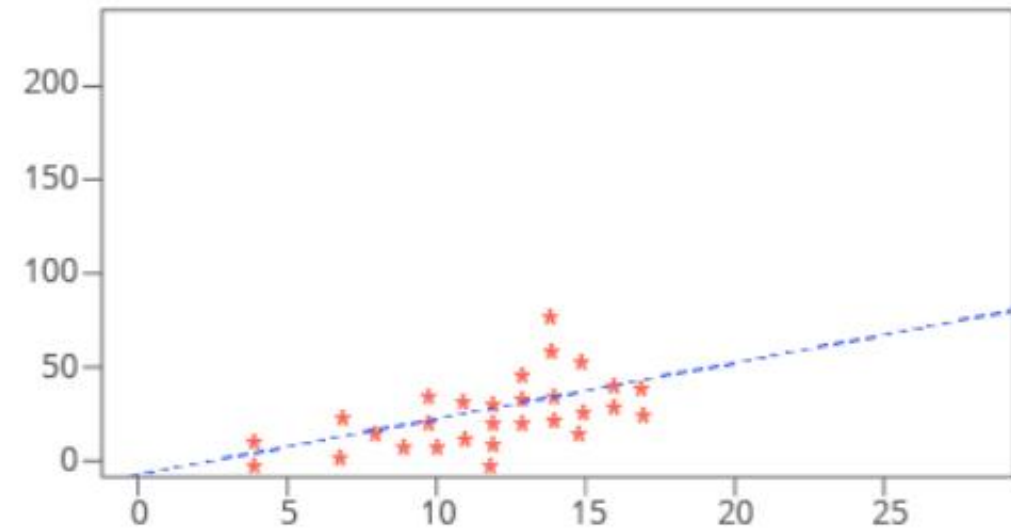
- Learning noise in the data
- Using a complex model for a simple task
- Inappropriate model hyperparameters

Reasons for Overfitting

Learning noise in the data



With outliers

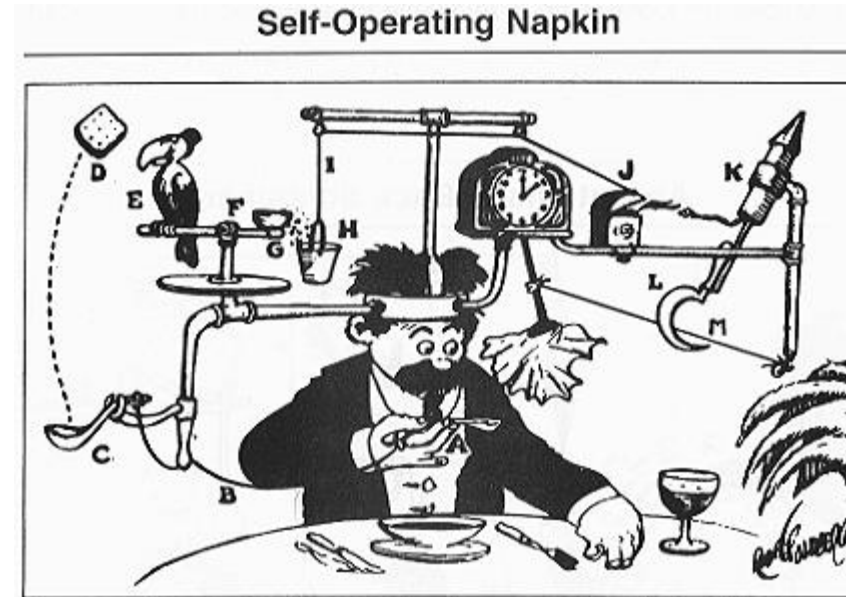


Without outliers

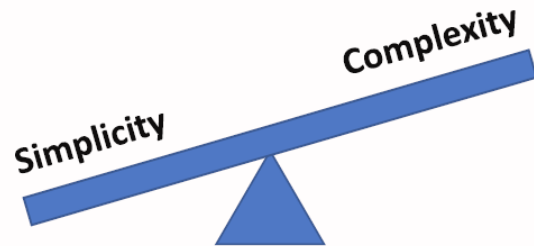
A much better fit

Reasons for Overfitting

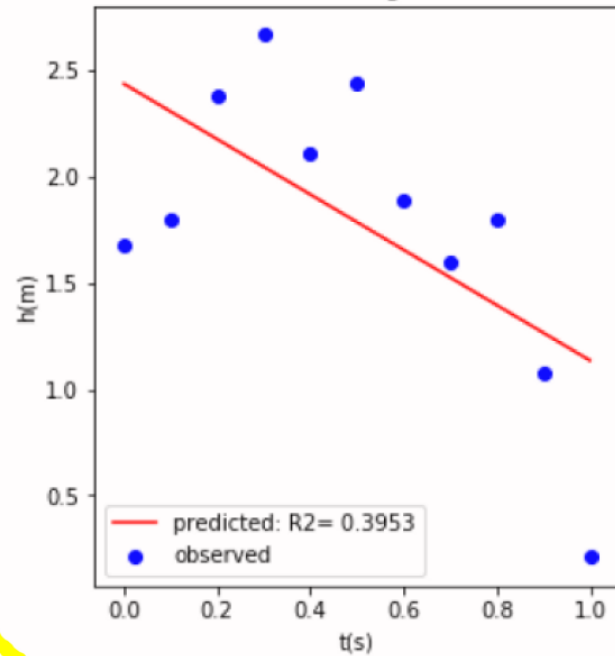
Using a complex algorithm for a simple task



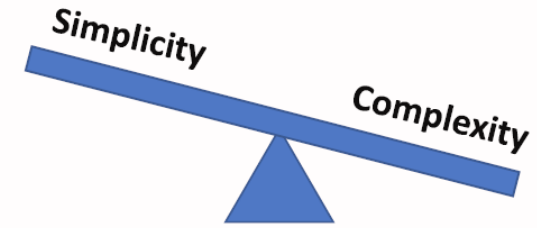
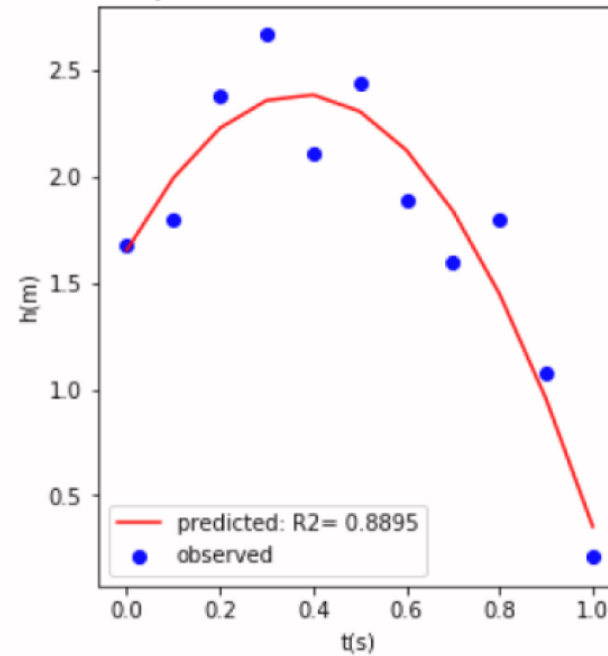
A **Rube Goldberg machine**, named after American cartoonist Rube Goldberg, is a machine intentionally designed to perform a simple task in an indirect and overly complicated way.



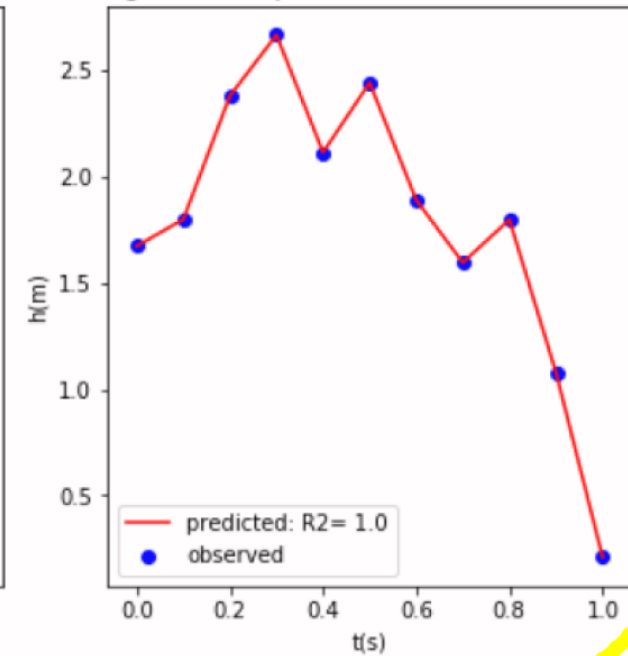
linear model is not good (underfit)



quadratic model is what we need



degree=10 captures random error (overfit)



Reasons for Overfitting

Inappropriate model hyperparameter(s)

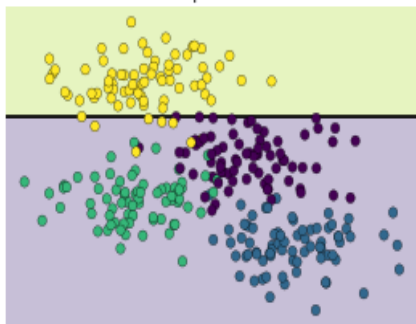


A simple decision tree built on this data will iteratively split the data along one or the other axis according to some quantitative criterion

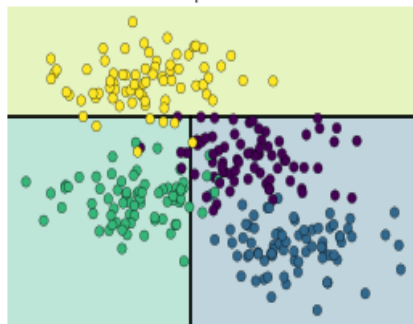


With default `max_depth = None`, the nodes are expanded until all leaves are pure

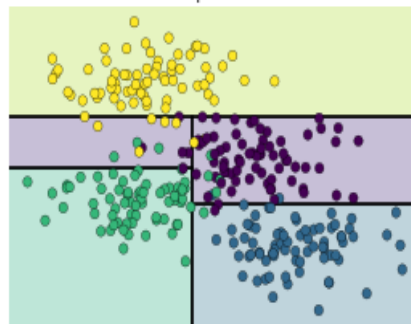
Depth = 1



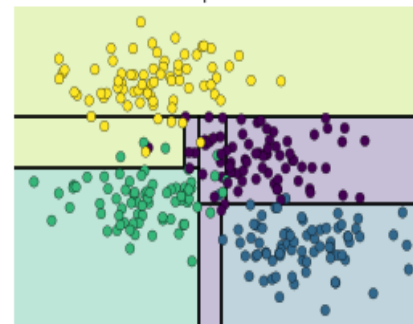
Depth = 2



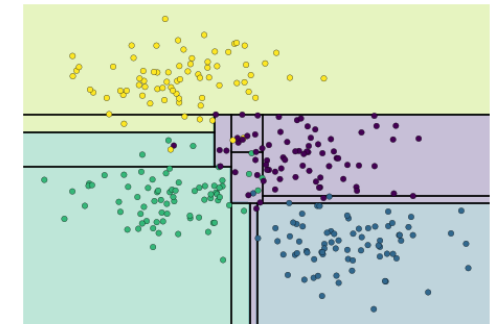
Depth = 3



Depth = 4



Depth = 5



.....

Overfitting Example

Hyperparameter tuning – Decision tree

- **Demo_Overfitting_Diabetes**

- Dataset: Diabetes dataset
- Objective: To understand the influence of tuning hyperparameters to avoid Overfitting

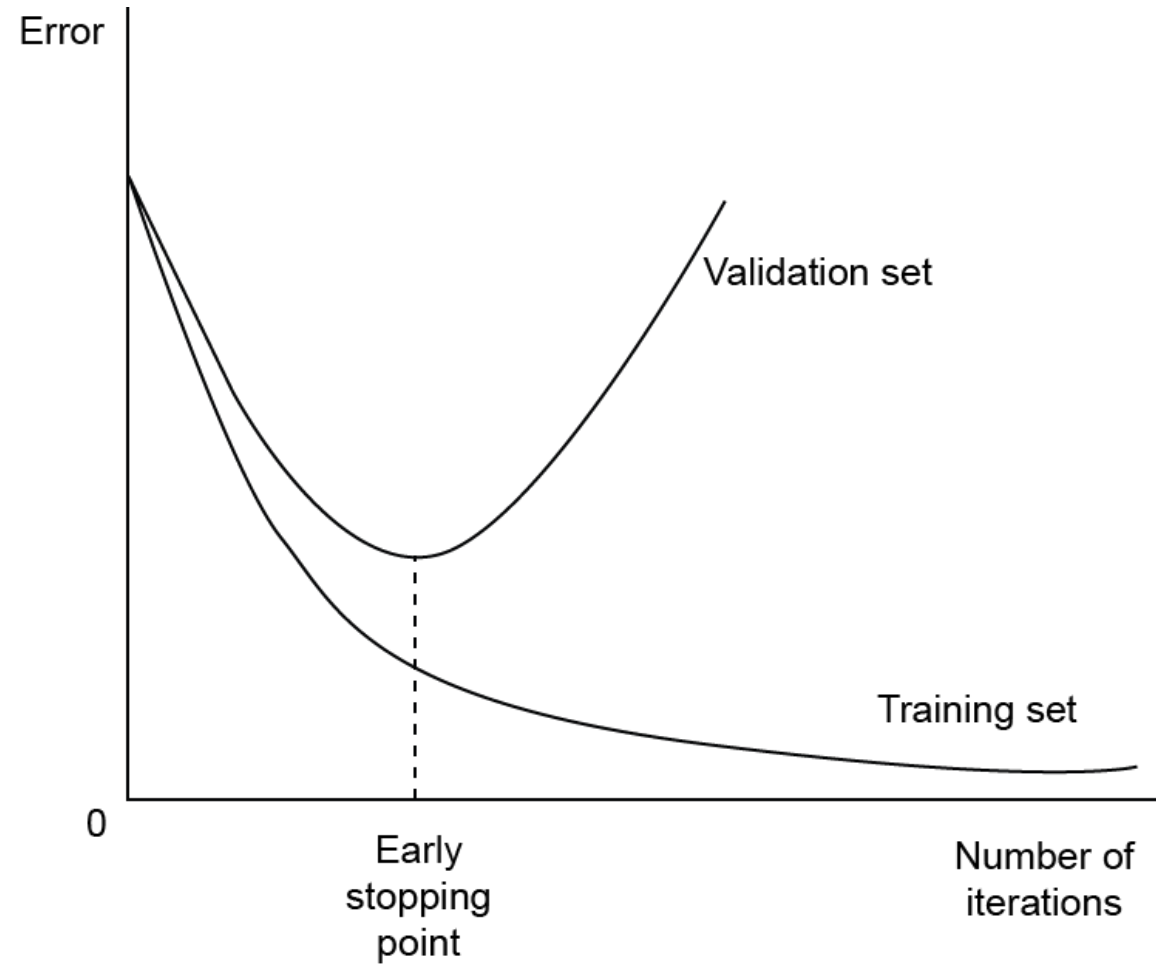
Addressing Overfitting

- Use More Data → Can learn more parameters
- Reduce the number of features or dimensions
 - Select which features to keep
 - Reduce dimension
 - Select which model to use

How to prevent/reduce overfitting?

- Train with more data
- Reduce/Remove features
- **Early Stopping**
- **Regularization**
- Ensembling

Early Stopping



Regularization

Regularization is a process of introducing additional information in order to solve an ill-posed problem or to prevent overfitting.

What to Minimize? (regularization)

- We were minimizing:

$$\varepsilon = \sum_i (z_i - y_i)^2 = \sum_i (z_i - f(x_i))^2$$

- We can minimize

Fit error + some measure of complexity of the model

ε + Sum of squares of Weights

ε + Number of Non-zero Weights

https://scikit-learn.org/stable/modules/generated/sklearn.linear_model.LinearRegression.html

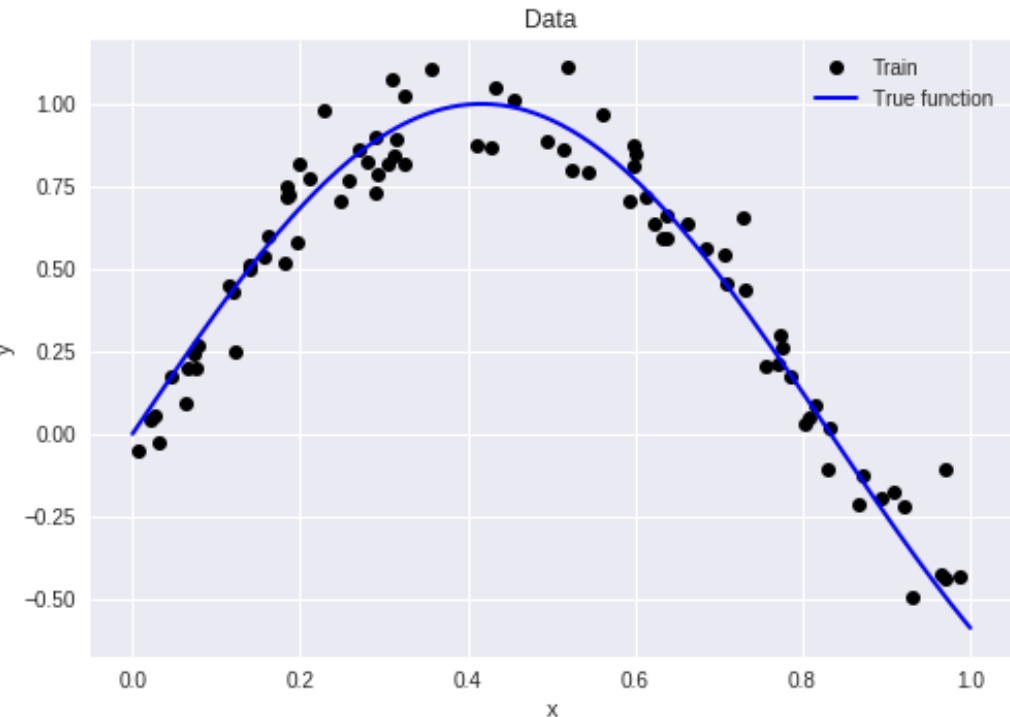
Ridge Regression (L2-Norm), Lasso Regression (L1-norm) - Least Absolute Shrinkage and Selection Operator

Effect of Regularization

Fit a curve to the data

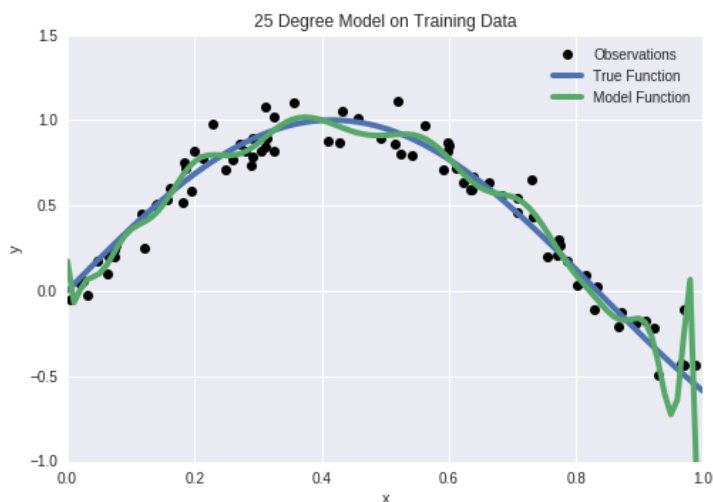
- Learn W to fit the following polyn regression

$$y = w_{25}.x^{25} + w_{24}.x^{24} + .. + w_1.x + w_0$$

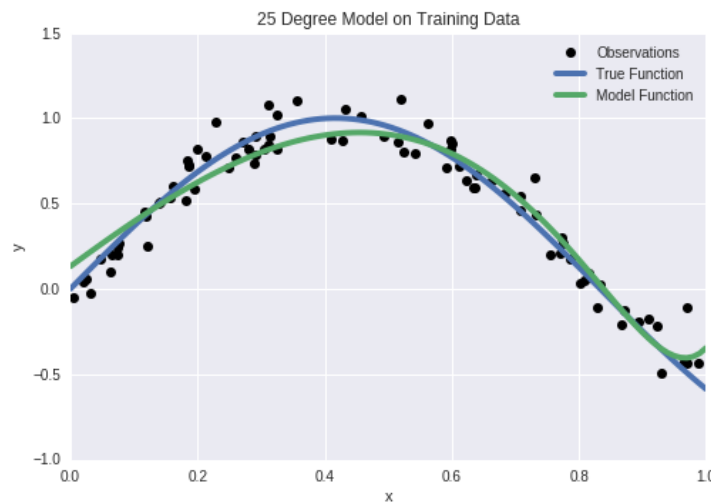


Blue curve represents the true function

Curve fitting with different regularizations



No Regularisation



L1 Regularisation
 $\lambda = 0.0125$



L2 Regularisation
 $\lambda = 0.0125$

Lambda=Regularization Parameter (alpha)

Summary

- Find model that is simple and fitting the data.
- Problem of overfitting.
 - How to detect? What to observe during learning?
 - How to avoid.
- Regularize the solution by adding extra term that also minimize the “complexity”.
- Right way to design solution with
 - Train, Val and Test splits.

Thanks!!

Questions?